

Problems Proposed for 2025-26 Pal4Pal

Alex, Edward, Kennan

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Alex

Q1 Combo [4 points]

Suppose Bob starts with 1 credit. Everyday Bob can choose to go to a casino. On Day i (with i starting from 1), he has a $\frac{101-i}{100}$ chance to win his gambling. If he wins, he wins a prize of $\frac{1}{2^i}$ credits. If he loses, then he loses all of the credits accumulated so far, and he has no more money to gamble anymore. Optimally, how many days of gambling should Bob do?

Q2 Number Theory [5 points]

A number a is n -good if it can be written as $r + bn$ where $0 < r < n$ and $\gcd(r, n) = 1, \gcd(b, n) = 1$. Find the size of the set of distinct remainders mod 3^{200} of perfect squares which are 3^{100} -good.

(*In other words, find $|\{a^2 \bmod 3^{200} | a^2 \text{ is } 3^{100}\text{-good}\}|$.)

Format for answering: If this quantity is expressed in the form of $a \times b^k$, find $a + b + k$.

Edward

Q1 Algebra [5 points]

Consider the sequence $\{a_n\}$, where a_n is the n th term in the sequence defined by $a_1 = 1, a_2 = 2$ and the recursive relation $a_n = 4a_{n-1} - 4a_{n-2} + 1$ for all $n \geq 3$.

It is possible to define a_n for all $n \leq 0$ requiring the recurrence relation to also be true for $n \leq 2$. In this way, all a_n for $n \leq 0$ can be written as $\frac{a}{2^b}$, where a is odd. Find the sum of the absolute values for the greatest (least negative e.g. $-5 > -7$) 3 n 's where the value of b for n and $n + 1$ differs by at least 9.

(For example: $n = -5$ does not satisfy this condition because $a_{-5} = \frac{61}{2^6}$ and $a_{-4} = \frac{59}{2^6}$, and the red and blue numbers (the 6's) do not differ by at least 9)

Q2 Number Theory [4 points]

Find the size (cardinality) of the largest subset $S \subseteq \{1, 2, 3, \dots, 100\}$ such that for any distinct $a, b, c \in S$, $\gcd(a, b, c) = 1$.

Kennan

Q1 Geo [4 points]

Suppose for a positive number x , we have a quadrilateral with sides $x, 2, 4 - x, 2$ in that order. It contains an inscribed circle of radius x with another inscribed quadrilateral in it that has corresponding parallel edges. Find x .

Q2 Combo [4 points]

Bob is walking in the cells of a 10×10 grid. Meanwhile, Alice is also walking in the cells of the same 10×10 grid. Alice starts from the bottom-left corner and Bob starts from the top-right corner. Alice walks up or right every minute randomly and Bob also randomly walks down or left every minute. Let the probability that they will collide be $\frac{p}{2^q}$ in simplest form and find q .